DiFrauD

(Domain-Independent Fraud Detection Benchmark)

DiFrauD (Domain-Independent Fraud Detection) is a large scale, multi-domain dataset designed to benchmark the deception detection systems across different fraud scenarios.

Findings:

1. Altogether there are 95,854 samples in the dataset across 7 different domains, where 37,282 texts are deceptive (38.9%) and 58,572 texts are non-deceptive (61.1%).

2. Out of 7 domains, Job Scams class had the highest class imbalance in which only 4.2% of the samples are deceptive.

Proposed Model:

1. I implemented a DistilBERT-based deep learning detector which primarily uses balanced sampling, weighted loss function and early stopping to handle data imbalances.

2. In-domain evaluation showed strong performance, whereas in Cross-domain evaluation, performance degrades for highly specialized domains.

DiFrauD Dataset Information:

In the dataset, each sample is a binary-labeled text where label = 1 represents deceptive content and label = 0 indicates non-deceptive content.

There are 7 different domains present in the dataset. They are:

1. Fake News (Deceptive: 8,832 & Non-Deceptive: 11,624)
2. Job Scams (Deceptive: 599 & Non-Deceptive: 13,696)
3. Phishing (Deceptive: 6,074 & Non-Deceptive: 9,198)
4. Political Statements (Deceptive: 8,042 & Non-Deceptive: 4,455)
5. Product Reviews (Deceptive: 10,492 & Non-Deceptive: 10,479)
6. SMS (Deceptive: 1,274 & Non-Deceptive: 5,300)
7. Twitter Rumors (Deceptive: 1,969 & Non-Deceptive: 3,820)

Data Cleaning in DiFrauD:

Data was curated and cleaned using Cleanlab and also the HTML tags, duplicate entries and non-English texts were removed. Similarly, in Job Scams domain, the label inversion bug from the original EMSCAD source was corrected. Whitespace, Unicode, quotation marks, and bullet characters were normalized as well.

Finally, each class was stored in JSONL format containing two fields viz. text and label & the dataset used the train, validation and text split of 80/10/10 with random seed of 42.

Exploratory Data Analysis (EDA):

1. The major challenge in training the model was the extreme class imbalance. The Job Scams domain had the highest imbalance as it contained only 4.2% deceptive samples.

2. Not only this, the text length across different domains was also diverse. For example: SMS had 10-50 words, Twitter Rumors had 20-100 words, Political Statements had 10-40 words, Product Reviews had 50-300 words, Phishing Emails had 100- 2000+ words, Fake News had 200-5000+ words and Job Scams had 100-3000+ words.

   Therefore, for tokenization, I used 256 tokens as a balance between coverage and computational cost.

3. If a model is trained on one type of fraud, in order to detect fraud in completely different domain, transformer models are better at capturing those deceptions as they understand the context and meaning instead of words.

4. There exists Linguistic Differences across different domains. Example: For Phishing emails, urgency words are used, suspicious links are present, and so on.

Potential Limitations of DiFrauD:

1. Since it contains the data only in the English language, its performance may not be consistent across different linguistics.

2. The data present in the dataset are all real-world data. So, its performance against the AI-generated fraud data is yet to be evaluated.

3. As the data present in the DiFrauD is of certain years' interval, its performance might not be the same in the recent and more fraudulent deceptive data.

Deep Learning Detector Design

I used DistilBERT as a deep learning detector because it retains approx. 97% of BERT's performance on downstream Natural Language Processing tasks and also it uses 40% fewer parameters.

The architecture contains:

1. DistilBERT encoder (6 transformer layers, 768 hidden dim, 12 attention heads)
2. Dropout (p = 0.1)
3. Linear classification head: 768→2 classification heads (deceptive / non-deceptive)
4. Cross-Entropy Loss.

Hyperparameters:

Base Model	DistilBERT – base - uncased
Max Token Length	256
Batch Size	32
Learning Rate	2e - 5
Warmup Ratio	10% of total steps
LR Scheduler	Linear with warmup
Max Epochs	5
Early Stopping	2 epochs
Weight Decay	0.01
Gradient Clipping	Max norm = 1.0
Optimizer	AdamW
Seed	42

Three ways of handling class imbalance:

1. Weighted Random Sampling

   While training the model, each mini-batch were created using WeightedRandomSampler that ensures each batch has equal representation from both classes.

2. Weighted Loss Function

   Weights are calculated as:

   $$W\_c = N / (2 * N\_c)$$

   where, N = Total Samples,
   N_c = No. of Samples in Class c.

3. Macro-F1 Early Stopping

   Macro-F1 evaluated performance on both classes, preventing the model from being selected based on only the majority-class accuracy.

Experimental Results:

The experiment was carried out across in-domain and cross-domain.

In-Domain Performance:

Domain	Accuracy	Macro F1	ROC-AUC
Phishing	0.942	0.934	0.981
Product Reviews	0.955	0.955	0.989
SMS	0.983	0.965	0.996
Fake News	0.937	0.936	0.977
Political Statements	0.740	0.706	0.818
Twitter Rumors	0.782	0.769	0.851
Job Scams	0.987	0.731	0.891
Average	0.904	0.857	0.929

I. SMS achieved the highest Macro F1 accuracy of 0.965 because they possess certain kind of linguistic patterns that is easily captured by the BERT model
II. Job Scams achieved the highest accuracy but low Macro F1 accuracy because of drastic class imbalance.
III. Political Statements achieved the lowest accuracy because political deception texts are a bit subtler than fraud and require the context and information to understand them.

Cross-Domain Performance (Zero-Shot):

Domains	Macro F1 (Cross)	Macro F1 (In-Domain)	Drop
Fake News	0.811	0.936	-0.125
Job Scams	0.644	0.731	-0.087
Phishing	0.855	0.934	-0.079
Political Statements	0.652	0.706	-0.054
Product Reviews	0.873	0.955	-0.082
SMS	0.901	0.965	-0.064
Twitter Rumors	0.704	0.769	-0.065

I. From the above results, SMS is the most transferable domain as it has the highest accuracy in cross domain performance. Fake News had the highest drop in performance because of the long form of news articles.

Discussion:

I. The DistilBERT fine-tuned model shows best performance for balanced domains like product reviews, phishing and SMS where fraud signals are consistent and easy to capture by the model.
II. Use of early stopping with Macro-F1 prevents the converging to trivially high-accuracy but low-recall models.
III. Job Scams is a difficult domain for fraud detection despite balancing it.
IV. Cross-domain performance drastically decreases for the domains like Fake News and Phishing, as they contain large form of texts.

Conclusion:

Therefore, the report presents an overview of DiFrauD dataset across 7 real-world domains. I performed Exploratory Data Analysis (EDA) in order to identify the key challenges in existing data and class imbalance was a major hurdle which I needed to tackle while training the deep model.

In order to solve all these issues, I implemented the DistilBERT-based deep learning detector using 3 strategies to handle class imbalances, viz. balanced random sampling, inverse-frequency weighted loss, and macro-F1 driven early stopping. Finally, evaluation was done under In-domain as well as Cross domain (zero-shot).

The transformer architecture based model was better at capturing deception texts of SMS and Product Reviews but demonstrated very poor performance in Political Statements and Job Scams.

Hence, in order to improve the performance further, we can shift towards using larger and more powerful models like RoBERTa and implement data augmentation strategies, which can provide us better results.

References:

Book:

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Paper:

- Boumber, D., Qachfar, F.Z., & Verma, R. (2024). Domain-Agonistic Adapter Architecture for Deception Detection: Extensive Evaluations with the DIFrauD Benchmark. *LREC-COLING 2024*. https://aclanthology.org/2024.lrec-main.468
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- Wolf, T. et al. (2020). Transformers: State-of-the-Art Natural Language Processing. EMNLP 2020. https://aclanthology.org/2020.emnlp-demos.6

Dataset:

- DiFrauD HuggingFace Dataset: https://huggingface.co/datasets/difraud/difraud

AI Tools:

- Gemini (Google) and Claude (Anthropic) were used for reference and assistance during the preparation of this report.